PYTHON PROGRAMMING LAB RECORD

AIM • PROCEDURE • FLOWCHART • PROGRAM • SAMPLE OUTPUT

Based on the 10 Python programs supplied.

Note: The sample inputs and outputs are illustrative runs. In Program 5, the stray trailing text “7, 55” in the supplied file was removed because it is not valid Python syntax.

Experiment 1: Student Scholarship Eligibility

Aim

To check whether each student is eligible for a scholarship based on aggregate marks, subject marks, attendance, and family income.

Procedure

1. Read the number of students.

2. For each student, read name, aggregate marks, marks in two subjects, attendance percentage, and family income.

3. Check whether aggregate marks are at least 75, both subject marks are at least 40, attendance is at least 75%, and family income is not more than Rs. 2,00,000.

4. Display whether the student is eligible or not eligible.

5. Repeat until all students are checked.

Flowchart

START → Read number of students → Repeat for each student: Read student details → Check all five eligibility conditions → [All

Program

```
n = int(input("Enter number of students: "))

for i in range(n):
    name = input("Enter student name: ")
    marks = int(input("Enter aggregate marks: "))
    s1 = int(input("Enter subject 1 marks: "))
    s2 = int(input("Enter subject 2 marks: "))
    attendance = int(input("Enter attendance percentage: "))
    income = int(input("Enter family income: "))

    if marks >= 75 and s1 >= 40 and s2 >= 40 and attendance >= 75 and income <= 200000:
        print(name, "is eligible for scholarship")
    else:
        print(name, "is not eligible for scholarship")
```

Sample Input

```
Number of students: 2
Student 1: Ravi, aggregate 80, subject 1 45, subject 2 50, attendance 85, income 150000
Student 2: Sita, aggregate 70, subject 1 45, subject 2 50, attendance 90, income 120000
```

Output

```
Ravi is eligible for scholarship
Sita is not eligible for scholarship
```

Experiment 2: Monthly Electricity Bill Calculation

Aim

To calculate monthly electricity bills using slab rates, fixed charges, and an adjustment, and to identify an unusual increase.

Procedure

1. Read the number of months.

2. For each month, read electricity units consumed.

3. Calculate the energy charge: Rs. 2 per unit up to 100 units; Rs. 3 per unit for units from 101 to 200; and Rs. 5 per unit above 200.

4. Add Rs. 50 fixed charge.

5. If the bill is above Rs. 1,000, add Rs. 100; otherwise subtract Rs. 20.

6. Store units and bills, then display each month's bill.

7. From the second month onward, display a warning if the current bill is more than 1.5 times the previous bill.

Flowchart

```
START → Read number of months → For each month read units → Select slab rate and calculate charge → Add Rs. 50 → Bill > Rs. 1
```

Program

```
n = int(input("Enter number of months: "))
units_list = []
bills = []

for i in range(n):
    units = int(input("Enter electricity units: "))

    if units <= 100:
        bill = units * 2
    elif units <= 200:
        bill = 100 * 2 + (units - 100) * 3
    else:
        bill = 100 * 2 + 100 * 3 + (units - 200) * 5

    bill = bill + 50

    if bill > 1000:
        bill = bill + 100
    else:
        bill = bill - 20

    units_list.append(units)
    bills.append(bill)

print("\nMonthly Electricity Bills")

for i in range(n):
    print("Month", i + 1, "Units:", units_list[i], "Bill:", bills[i])

    if i > 0 and bills[i] > bills[i - 1] * 1.5:
        print("Unusual increase in bill")
```

Sample Input

```
Number of months: 3
Units for month 1: 80
Units for month 2: 150
Units for month 3: 250
```

Output

```
Monthly Electricity Bills
Month 1 Units: 80 Bill: 190
Month 2 Units: 150 Bill: 380
Unusual increase in bill
Month 3 Units: 250 Bill: 780
Unusual increase in bill
```

Experiment 3: Hospital Bill and Payment Category

Aim

To calculate a patient's hospital bill, apply a discount when eligible, and classify the final payment.

Procedure

1. Read the patient's name and consultation, diagnostic, and treatment charges.
2. Add the three charges to find the total.
3. If any charge is negative, display Invalid charges.
4. If the total is at least Rs. 5,000, calculate a 10% discount; otherwise, the discount is zero.
5. Subtract the discount from the total to get the final bill.
6. Classify the payment as High (Rs. 5,000 or more), Medium (Rs. 2,000 or more), or Low.
7. Display the patient name, total, discount, and final bill.

Flowchart

```
START → Read patient name and charges → Calculate total → Any charge negative? Yes: Display Invalid charges → STOP / No: Total
```

Program

```
name = input("Enter patient name: ")

c = int(input("Enter consultation charges: "))
d = int(input("Enter diagnostic charges: "))
t = int(input("Enter treatment charges: "))

total = c + d + t

if c < 0 or d < 0 or t < 0:
    print("Invalid charges")
else:
    if total >= 5000:
        discount = total * 10 / 100
    else:
        discount = 0

    bill = total - discount

    if bill >= 5000:
        print("High Payment")
    elif bill >= 2000:
        print("Medium Payment")
    else:
        print("Low Payment")

    print("Patient Name:", name)
    print("Total Bill:", total)
    print("Discount:", discount)
    print("Final Bill:", bill)
```

Sample Input

```
Patient name: Anil
Consultation charges: 2000
Diagnostic charges: 1500
Treatment charges: 2500
```

Output

```
High Payment
Patient Name: Anil
Total Bill: 6000
Discount: 600.0
Final Bill: 5400.0
```

Experiment 4: Employee Salary Statement

Aim

To calculate an employee's gross salary, deduction, and net salary.

Procedure

1. Read employee name, basic pay, allowances, overtime, and incentives.

2. Add all salary components to calculate gross salary.

3. If gross salary is up to Rs. 10,000, deduct 5%.

4. If gross salary is above Rs. 10,000 and up to Rs. 20,000, deduct 10%.

5. If gross salary is above Rs. 20,000, deduct 15%.

6. Subtract the deduction from gross salary to calculate net salary.

7. Display the salary statement.

Flowchart

```
START → Read employee and salary details → Calculate gross salary → Gross ≤ 10,000? 5% deduction : Gross ≤ 20,000? 10% deductio
```

Program

```
name = input("Enter employee name: ")
basic = float(input("Enter basic pay: "))
allowances = float(input("Enter allowances: "))
overtime = float(input("Enter overtime amount: "))
incentives = float(input("Enter incentives: "))

gross = basic + allowances + overtime + incentives

if gross <= 10000:
    deduction = gross * 0.05
elif gross <= 20000:
    deduction = gross * 0.10
else:
    deduction = gross * 0.15

net = gross - deduction

print("\n----- SALARY STATEMENT -----")
print("Employee Name:", name)
print("Basic Pay:", basic)
print("Allowances:", allowances)
print("Overtime:", overtime)
print("Incentives:", incentives)
print("Gross Salary:", gross)
print("Deduction:", deduction)
print("Net Salary:", net)
```

Sample Input

```
Employee name: Kiran
Basic pay: 12000
Allowances: 2000
Overtime: 500
Incentives: 500
```

Output

```
----- SALARY STATEMENT -----
Employee Name: Kiran
Basic Pay: 12000.0
Allowances: 2000.0
Overtime: 500.0
Incentives: 500.0
Gross Salary: 15000.0
Deduction: 1500.0
Net Salary: 13500.0
```

Experiment 5: Mobile Data Plan Selection

Aim

To select a suitable mobile data plan based on data usage, validity days, and the user's budget.

Procedure

1. Read required data usage, validity days, and budget.

2. Check whether Plan 1 meets the requirements: usage up to 2 GB, validity up to 28 days, and budget at least Rs. 199.

3. If not, check Plan 2: usage up to 3 GB, validity up to 28 days, and budget at least Rs. 299.

4. If not, check Plan 3: usage up to 2 GB, validity up to 56 days, and budget at least Rs. 399.

5. Display the first suitable plan and its cost, or display No suitable plan.

Flowchart

```
START → Read usage, validity, budget → Plan 1 conditions satisfied? Yes: Display Plan 1 / No: Plan 2 conditions satisfied? Yes:
```

Program

```
usage = int(input("Enter data usage: "))
days = int(input("Enter validity days: "))
budget = int(input("Enter budget: "))

if usage <= 2 and days <= 28 and budget >= 199:
    print("Plan 1 is best")
    print("Cost = 199")

elif usage <= 3 and days <= 28 and budget >= 299:
    print("Plan 2 is best")
    print("Cost = 299")

elif usage <= 2 and days <= 56 and budget >= 399:
    print("Plan 3 is best")
    print("Cost = 399")

else:
    print("No suitable plan")
```

Sample Input

```
Data usage: 2
Validity days: 28
Budget: 299
```

Output

```
Plan 1 is best
Cost = 199
```

Experiment 6: Ticket Fare and Cancellation Refund

Aim

To calculate a ticket fare based on distance and age, add a reservation charge, and calculate a refund if the ticket is cancelled.

Procedure

1. Read the travel distance and passenger age.

2. If distance is up to 100, calculate fare at Rs. 2 per unit of distance; otherwise calculate at Rs. 3 per unit.

3. If age is below 12, reduce the fare by half.

4. Add a reservation charge of Rs. 50.

5. Display the ticket fare.

6. Ask whether the ticket should be cancelled.

7. If the answer is yes, calculate an 80% refund; otherwise display Ticket Confirmed.

Flowchart

```
START → Read distance and age → Distance ≤ 100? Calculate at Rs. 2/unit : Calculate at Rs. 3/unit → Age < 12? Halve fare if yes
```

Program

```
distance = int(input("Enter distance: "))
age = int(input("Enter age: "))

if distance <= 100:
    fare = distance * 2
else:
    fare = distance * 3

if age < 12:
    fare = fare / 2

reservation = 50
total = fare + reservation

print("Ticket Fare:", total)

cancel = input("Cancel ticket? yes/no: ")

if cancel == "yes":
    refund = total * 0.8
    print("Refund Amount:", refund)
else:
    print("Ticket Confirmed")
```

Sample Input

```
Distance: 120
Age: 10
Cancel ticket? yes
```

Output

```
Ticket Fare: 230.0
Refund Amount: 184.0
```

Experiment 7: Shopping Bill with Discount and Tax

Aim

To calculate the total shopping amount, apply a discount with a maximum limit, and add tax to find the final bill.

Procedure

1. Read the number of products.

2. For each product, read its price and quantity.

3. Multiply price by quantity and add the amount to the running total.

4. If the total is at least Rs. 1,000, calculate a 10% discount; otherwise, discount is zero.

5. Limit the discount to a maximum of Rs. 200.

6. Subtract the discount, calculate 5% tax, and add it to get the final bill.

7. Display total amount, discount, tax, and final bill.

Flowchart

```
START → Read number of products → For each product read price and quantity → Calculate item amount and add to total → Total ≥
```

Program

```
n = int(input("Enter number of products: "))

total = 0

for i in range(n):
    price = float(input("Enter product price: "))
    qty = int(input("Enter quantity: "))

    amount = price * qty
    total = total + amount

if total >= 1000:
    discount = total * 0.10
else:
    discount = 0

if discount > 200:
    discount = 200

amount = total - discount
tax = amount * 0.05
bill = amount + tax

print("Total Amount:", total)
print("Discount:", discount)
print("Tax:", tax)
print("Final Bill:", bill)
```

Sample Input

```
Number of products: 2
Product 1 price: 600, quantity: 1
Product 2 price: 500, quantity: 2
```

Output

```
Total Amount: 1600.0
Discount: 160.0
Tax: 72.0
Final Bill: 1512.0
```

Experiment 8: Production Cost and Profit Target

Aim

To calculate the total production cost and profit, and check whether the profit target is achieved.

Procedure

1. Read material, labour, wastage, and machine costs.
2. Read the selling price and required profit target.
3. Add all costs to calculate total cost.
4. Subtract total cost from selling price to calculate profit.
5. Compare profit with the target.
6. Display whether the profit target is achieved.

Flowchart

```
START → Read all costs, selling price, and target → Calculate total cost → Calculate profit = selling price − total cost → Pro
```

Program

```
material = float(input("Enter material cost: "))
labour = float(input("Enter labour cost: "))
wastage = float(input("Enter wastage cost: "))
machine = float(input("Enter machine cost: "))
selling = float(input("Enter selling price: "))
target = float(input("Enter profit target: "))

cost = material + labour + wastage + machine
profit = selling - cost

print("Total Cost:", cost)
print("Profit:", profit)

if profit >= target:
    print("Profit Target Achieved")
else:
    print("Profit Target Not Achieved")
```

Sample Input

```
Material cost: 500
Labour cost: 200
Wastage cost: 50
Machine cost: 100
Selling price: 1200
Profit target: 300
```

Output

```
Total Cost: 850.0
Profit: 350.0
Profit Target Achieved
```

Experiment 9: Travel Expense and Budget Analysis

Aim

To calculate travel expenses in rupees after currency conversion and fee, check the budget, and identify the largest expense category.

Procedure

1. Read hotel, food, and travel expenses in foreign currency.

2. Read the exchange rate, conversion fee percentage, and rupee budget.

3. Add the three expenses and multiply by the exchange rate.

4. Calculate the conversion fee percentage and add it to the total.

5. Compare the total expense with the budget and display the result.

6. Compare hotel, food, and travel expenses to identify the largest category.

Flowchart

```
START → Read hotel, food, travel, exchange rate, fee, and budget → Convert sum of expenses into rupees → Calculate and add conv
```

Program

```
hotel = float(input("Enter hotel expense: "))
food = float(input("Enter food expense: "))
travel = float(input("Enter travel expense: "))

rate = float(input("Enter exchange rate: "))
fee = float(input("Enter conversion fee percentage: "))
budget = float(input("Enter budget in rupees: "))

total = (hotel + food + travel) * rate
conversion_fee = total * fee / 100
total = total + conversion_fee

print("Total Expense:", total)
print("Conversion Fee:", conversion_fee)

if total <= budget:
    print("Within Budget")
else:
    print("Over Budget")

if hotel >= food and hotel >= travel:
    print("Largest Expense: Hotel")
elif food >= travel:
    print("Largest Expense: Food")
else:
    print("Largest Expense: Travel")
```

Sample Input

```
Hotel: 100
Food: 50
Travel: 30
Exchange rate: 80
Conversion fee percentage: 2
Budget in rupees: 15000
```

Output

```
Total Expense: 14688.0
Conversion Fee: 288.0
Within Budget
Largest Expense: Hotel
```

Experiment 10: Event Budget and Minimum Participants

Aim

To calculate event costs, sponsorship shortfall, minimum participants required, and participant contribution.

Procedure

1. Read venue, equipment, catering, transport, and certificate cost per person.

2. Read sponsorship amount and registration fee per person.

3. Add venue, equipment, catering, and transport costs to get fixed cost.

4. Subtract sponsorship from fixed cost to get the shortfall. If it is negative, set it to zero.

5. Calculate the number of participants using ceiling division of shortfall by registration fee.

6. Add certificate costs for all participants to fixed cost.

7. Subtract sponsorship from total cost to calculate participant contribution.

8. Display all calculated values.

Flowchart

```
START → Read fixed costs, certificate cost, sponsorship, and registration fee → Calculate fixed cost → Calculate shortfall = fi
```

Program

```
venue = int(input("Enter venue cost: "))
equipment = int(input("Enter equipment cost: "))
catering = int(input("Enter catering cost: "))
transport = int(input("Enter transport cost: "))
certificate = int(input("Enter certificate cost per person: "))

sponsor = int(input("Enter sponsorship amount: "))
fee = int(input("Enter registration fee per person: "))

cost = venue + equipment + catering + transport

remaining = cost - sponsor

if remaining < 0:
    remaining = 0

people = remaining // fee

if remaining % fee != 0:
    people = people + 1

total = cost + certificate * people
contribution = total - sponsor

print("Total Fixed Cost:", cost)
print("Sponsorship Shortfall:", remaining)
print("Minimum Participants:", people)
print("Total Cost:", total)
print("Participant Contribution:", contribution)
```

Sample Input

```
Venue: 10000
Equipment: 5000
Catering: 8000
Transport: 2000
Certificate cost per person: 100
Sponsorship: 5000
Registration fee per person: 500
```

Output

```
Total Fixed Cost: 25000
Sponsorship Shortfall: 20000
Minimum Participants: 40
Total Cost: 29000
Participant Contribution: 24000
```